

1 (a) (i) $(50 \text{ to } 200) \times 10^{-3} \text{ kg}$ or $(0.05 \text{ to } 0.2) \text{ kg}$ B1 [1]

(ii) $(50 \text{ to } 300) \text{ cm}^3$ B1 [1]

(b) density = mass/volume or $\rho = M/V$ C1

$$V = [\pi(0.38 \times 10^{-3})^2 \times 25.0 \times 10^{-2}]/4 (= 2.835 \times 10^{-8} \text{ m}^3) \quad \text{C1}$$

$$\begin{aligned} \rho &= (0.225 \times 10^{-3})/2.835 \times 10^{-8} \\ &= 7940 \text{ (kg m}^{-3}\text{)} \end{aligned} \quad \text{A1}$$

$$\Delta\rho/\rho = 2(0.01/0.38) + (0.1/25.0) + (0.001/0.225) [= 0.061]$$

or

$$\%\rho = 5.3\% + 0.40\% + 0.44\% (= 6.1\%) \quad \text{C1}$$

$$\Delta\rho = 0.061 \times 7940 = 480 \text{ (kg m}^{-3}\text{)}$$

density = $(7.9 \pm 0.5) \times 10^3 \text{ kg m}^{-3}$ or $(7900 \pm 500) \text{ kg m}^{-3}$ A1 [5]